

Unit 41: Improper Integrals — Full Solutions

Warm-up

1. $\int_1^{\infty} \frac{1}{x^2} dx$

$$\lim_{b \rightarrow \infty} \int_1^b x^{-2} dx = \lim_{b \rightarrow \infty} \left[-x^{-1} \right]_1^b = \lim_{b \rightarrow \infty} \left(-\frac{1}{b} + 1 \right) = 0 + 1 = 1$$

Converges to 1.

2. $\int_1^{\infty} \frac{1}{x} dx$

$$\lim_{b \rightarrow \infty} \left[\ln x \right]_1^b = \lim_{b \rightarrow \infty} (\ln b - 0) = \infty$$

Diverges.

3. $\int_0^{\infty} e^{-x} dx$

$$\lim_{b \rightarrow \infty} \left[-e^{-x} \right]_0^b = \lim_{b \rightarrow \infty} (-e^{-b} + 1) = 0 + 1 = 1$$

Converges to 1.

4. $\int_{-\infty}^0 e^x dx$

$$\lim_{a \rightarrow -\infty} \left[e^x \right]_a^0 = \lim_{a \rightarrow -\infty} (1 - e^a) = 1 - 0 = 1$$

Converges to 1.

5. $\int_1^{\infty} \frac{1}{x^3} dx$

$$\lim_{b \rightarrow \infty} \left[-\frac{1}{2x^2} \right]_1^b = \lim_{b \rightarrow \infty} \left(-\frac{1}{2b^2} + \frac{1}{2} \right) = 0 + \frac{1}{2} = \frac{1}{2}$$

Converges to $\frac{1}{2}$.

6. $\int_4^{\infty} \frac{1}{x^{3/2}} dx$

$$\lim_{b \rightarrow \infty} \left[-2x^{-1/2} \right]_4^b = \lim_{b \rightarrow \infty} \left(-\frac{2}{\sqrt{b}} + \frac{2}{\sqrt{4}} \right) = 0 + 1 = 1$$

Converges to 1.

Standard

$$7. \int_0^1 \frac{1}{x} dx$$

Discontinuity at $x = 0$.

$$\lim_{a \rightarrow 0^+} \left[\ln x \right]_a^1 = \lim_{a \rightarrow 0^+} (0 - \ln a) = \infty$$

Diverges.

$$8. \int_0^1 \frac{1}{x^2} dx$$

$$\lim_{a \rightarrow 0^+} \left[-x^{-1} \right]_a^1 = \lim_{a \rightarrow 0^+} \left(-1 + \frac{1}{a} \right) = \infty$$

Diverges.

$$9. \int_0^4 \frac{1}{\sqrt{x}} dx$$

$$\lim_{a \rightarrow 0^+} \left[2x^{1/2} \right]_a^4 = \lim_{a \rightarrow 0^+} (4 - 2\sqrt{a}) = 4 - 0 = 4$$

Converges to 4.

$$10. \int_{-\infty}^{\infty} \frac{1}{1+x^2} dx$$

Split at 0.

$$\int_0^{\infty} \frac{dx}{1+x^2} = \lim_{b \rightarrow \infty} \left[\arctan x \right]_0^b = \frac{\pi}{2} - 0 = \frac{\pi}{2}$$

$$\int_{-\infty}^0 \frac{dx}{1+x^2} = \lim_{a \rightarrow -\infty} \left[\arctan x \right]_a^0 = 0 - \left(-\frac{\pi}{2} \right) = \frac{\pi}{2}$$

Both converge, so the total:

$$\frac{\pi}{2} + \frac{\pi}{2} = \pi$$

Converges to π .

$$11. \int_2^{\infty} \frac{1}{(x-1)^2} dx$$

No discontinuity on $[2, \infty)$ (since $x-1 \geq 1 > 0$ throughout).

$$\lim_{b \rightarrow \infty} \left[-\frac{1}{x-1} \right]_2^b = \lim_{b \rightarrow \infty} \left(-\frac{1}{b-1} + 1 \right) = 0 + 1 = 1$$

Converges to 1.

$$12. \int_0^3 \frac{1}{3-x} dx$$

Discontinuity at $x = 3$ (upper endpoint). Antiderivative: $-\ln|3-x|$.

$$\lim_{b \rightarrow 3^-} \left[-\ln|3-x| \right]_0^b = \lim_{b \rightarrow 3^-} (-\ln(3-b) + \ln 3)$$

As $b \rightarrow 3^-$, $(3-b) \rightarrow 0^+$, so $\ln(3-b) \rightarrow -\infty$, meaning $-\ln(3-b) \rightarrow +\infty$.

Diverges.

Challenge

$$13. \int_0^\infty \frac{1}{1+x^2} dx$$

$$\lim_{b \rightarrow \infty} \left[\arctan x \right]_0^b = \lim_{b \rightarrow \infty} \arctan b - 0 = \frac{\pi}{2}$$

Converges to $\frac{\pi}{2}$.

$$14. \int_1^\infty \frac{1}{x^{3/2}} dx$$

$$\lim_{b \rightarrow \infty} \left[-2x^{-1/2} \right]_1^b = \lim_{b \rightarrow \infty} \left(-\frac{2}{\sqrt{b}} + 2 \right) = 0 + 2 = 2$$

Converges to 2.

$$15. \int_0^1 \frac{1}{\sqrt{x}} dx$$

$$\lim_{a \rightarrow 0^+} \left[2x^{1/2} \right]_a^1 = \lim_{a \rightarrow 0^+} (2 - 2\sqrt{a}) = 2 - 0 = 2$$

Converges to 2.

$$16. \int_{-1}^1 \frac{1}{x^{3/4}} dx$$

A note on interpretation: $x^{3/4}$ is not real-valued for negative x under the standard definition. Reading this integral as intended to be evaluated over the full symmetric interval $[-1, 1]$, we interpret the integrand as $\frac{1}{|x|^{3/4}}$, which is real-valued and even.

Since the integrand is even, and there's a discontinuity at $x = 0$:

$$\int_{-1}^1 \frac{dx}{|x|^{3/4}} = 2 \int_0^1 x^{-3/4} dx$$

$$= 2 \lim_{a \rightarrow 0^+} \left[4x^{1/4} \right]_a^1 = 2 \lim_{a \rightarrow 0^+} (4 - 4a^{1/4}) = 2(4 - 0) = 8$$

Converges to 8 (under the $|x|^{3/4}$ interpretation).

17. $\int_{-2}^2 \frac{1}{x^2} dx$

This interval has a hidden discontinuity at the interior point $x = 0$ — easy to miss since 0 isn't an endpoint! We must split there:

$$\int_0^2 \frac{dx}{x^2} = \lim_{a \rightarrow 0^+} \int_a^2 x^{-2} dx = \lim_{a \rightarrow 0^+} [-x^{-1}]_a^2 = \lim_{a \rightarrow 0^+} \left(-\frac{1}{2} + \frac{1}{a} \right) = \infty$$

Since this piece alone already diverges, the entire original integral diverges — there's no need to even check the other half (though by symmetry, it would diverge too).

Diverges.

Applied

18. $D(t) = D_0 e^{-kt}$

$$\int_0^\infty D_0 e^{-kt} dt = \lim_{b \rightarrow \infty} \left[-\frac{D_0}{k} e^{-kt} \right]_0^b = \lim_{b \rightarrow \infty} \left(-\frac{D_0}{k} e^{-kb} + \frac{D_0}{k} \right) = 0 + \frac{D_0}{k} = \frac{D_0}{k}$$

General formula: $\frac{D_0}{k}$.

With $D_0 = 100$, $k = 0.2$:

$$\frac{100}{0.2} = 500$$

Answer: total accumulated dose = 500 mg.

19. $\int_0^\infty 2e^{-2x} dx$

$$\lim_{b \rightarrow \infty} \left[-e^{-2x} \right]_0^b = \lim_{b \rightarrow \infty} (-e^{-2b} + 1) = 0 + 1 = 1$$

Confirmed: the integral equals exactly 1, verifying this is a valid total-probability distribution.

20. $\int_0^\infty 500e^{-0.05t} dt$

Using the same pattern as Problem 18, with $D_0 = 500$, $k = 0.05$:

$$\frac{500}{0.05} = 10,000$$

Answer: present value = \$10,000.

21. $\int_0^1 \frac{1}{\sqrt{1-x}} dx$

Discontinuity at $x = 1$ (upper endpoint). Let $u = 1 - x$, $du = -dx$:

$$\int \frac{1}{\sqrt{1-x}} dx = \int u^{-1/2}(-du) = -2u^{1/2} + C = -2\sqrt{1-x} + C$$

$$\lim_{b \rightarrow 1^-} \left[-2\sqrt{1-x} \right]_0^b = \lim_{b \rightarrow 1^-} (-2\sqrt{1-b} + 2)$$

As $b \rightarrow 1^-$, $(1-b) \rightarrow 0^+$, so $\sqrt{1-b} \rightarrow 0$:

$$= 0 + 2 = 2$$

Answer: total work = 2 (converges, despite the force growing without bound at $x = 1$).